Unit 25 Java™

1 *Java applets*

A Match the examples of Java programs, known as *applets*, (a–e) with the descriptions (1–5).

1 This Land Rover applet allows you to change the look of the vehicle.

2 The Pythagoras theorem applet gives the proof of the Pythagorean theorem without words. It allows you to manipulate triangles and go through the steps of the geometrical proof.

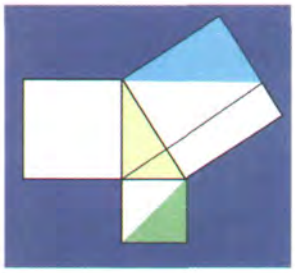
3 The Jman for Java applet permits medical researchers to view sequential MRI (**M**agnetic **R**esonance **I**mages) of the brain.

4 An analogue clock applet displays the time according to the web user's computer and lets you set the colours and style of the hands and numbers.

5 A banner applet displays graphic images on websites in order to advertise products or services.

a 4

c 3



e 2

b 1

d 5

B Match the terms (1–5) with the definitions (a–e).

1 Java C

2 applet E

3 plug-in A

4 platform-independent B

5 object-oriented programming D

a an auxiliary program that enables web browsers to support new content, for example animation

b software that can run on any operating system

c an island in Indonesia, coffee (in American slang), and a programming language for internet applications

d a computer programming technique that allows the creation of objects that interact with each other and can be used as the foundation of others; used to create graphical user interfaces

e a small Java application, usually designed to run automatically within a web page

The Java logo

2 The Java language

A These statements about Java are all false. Read the text and correct them.

1 Java was invented by Microsoft. by Sun Microsystems.
2 With the interpreter, a program is first converted into Java bytecodes.
3 Java is not compatible with most computing platforms.
4 The Java language is single-threaded, one part executing at a time.
5 Java has no competitors.
6 Flash files are called *animations*.

The Java language

Java is a programming language developed by Sun Microsystems, specially designed to run on the Web. Java programs (called **applets**) let you watch animated characters and moving text, play music, and interact with information on the screen (for example, control animations and select options).

Characteristics of the Java language

Java is an **object-oriented** language, similar to C++, but more dynamic and simplified to eliminate possible programming errors. A Java program is both compiled and interpreted (see Unit 24). First, the source code (a file with a **.java** extension) is compiled and converted into a format called bytecode (a file with a **.class** extension), which can then be executed by a Java interpreter (see Fig. 1). Compiled Java code can run on most computers because there are Java interpreters, known as **Java Virtual Machines**, for most operating systems.

Java is **multi-threaded**, meaning a Java program can have multiple threads (parts) – that is, many different things processing independently and continuously. This enables the program to make the best use of available CPU power.

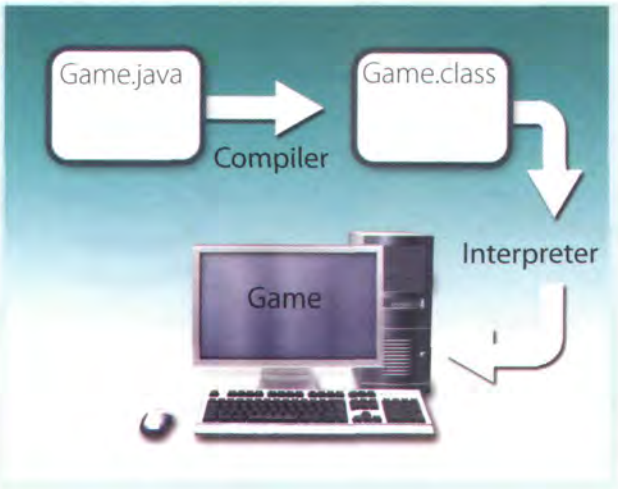


Fig. 1

Why is Java popular?

Most programmers like Java because it allows them to write applets which make web pages more interactive and attractive. They can create graphical objects (for example, bar charts and diagrams) and new controls (for example, check boxes and push buttons with special properties). A web page that uses Java can have sounds that play in real time, music that plays in the background, cartoon-style animations, real-time video and interactive games.

The Java Micro Edition platform (**Java ME**) is used in mobile devices. It provides flexible tools to create applications that run on mobile phones, PDAs, TV set-top boxes and printers. Nowadays, most phones are configured to use Java games.

Alternatives to Java

One alternative to Java is Microsoft's **C#**, pronounced 'C sharp', a **.NET** language based on C++ with elements from Visual Basic and Java. There are no substantial differences between C# and Java. When software developers do measurements on pieces of code, sometimes Java is faster, sometimes C# is.

Another competitor is Adobe **Flash** technology, which supports graphics, a scripting language called ActionScript, and the streaming of audio and video. Flash is used to create animation and advertisements, to integrate video into web pages, and to develop rich internet applications such as portals. **Flash files**, traditionally called **flash movies**, have a **.swf** file extension. They may be an object on a web page or be played in the stand-alone Flash Player.